Base

$$
\begin{array}{rcl}
\tau & ::= & o \\
& | & \tau \to \tau \\
& | & \langle \tau, \tau \rangle \\
& | & \mathsf{unit} \\
\\
o & ::= & \mathrm{num} \\
& | & \mathrm{bool} \\
\\
e & ::= & n \mid b \mid \mathsf{unit} \mid \langle e, e \rangle \mid x \\
& | & \lambda x : \tau . e \\
& | & e + e \mid e - e \mid e \wedge e \mid e \vee e \\
& | & e\, e \mid \mathrm{if}\ e\ \mathrm{then}\ e\ \mathrm{else}\ e \\
& | & \mathsf{zero?}(e) \\
& | & \mathsf{inj.l}(e) \mid \mathsf{inj.r}(e)
\end{array}
$$

$$
\begin{array}{rcl}
n & ::= & 0 \mid -1 \mid 1 \mid \ldots \\
\\
b & ::= & \mathsf{true} \mid \mathsf{false} \\
\\
v & ::= & n \mid b \mid \mathsf{unit} \mid \langle v, v \rangle \mid x \mid \lambda x : \tau . e \\
\\
E & ::= & \square \\
& | & E + e \mid v + E \mid E - e \mid v - E \\
& | & E \wedge e \mid v \wedge E \mid E \vee e \mid v \vee E \\
& | & E\, e \mid v\, E \mid \mathrm{if}\ E\ \mathrm{then}\ e\ \mathrm{else}\ e \mid \mathsf{zero?}(E) \\
& | & \langle E, e \rangle \mid \langle v, E \rangle, \mathsf{inj.l}(E) \mid \mathsf{inj.r}(E)
\end{array}
$$

Figure 1: Base Language